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AI-Transformed Air Quality Monitoring from Space

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Abstract

The application of artificial intelligence (AI) has been rapidly expanding in the field of remote sensing. However, most AI-based remote sensing studies are data-driven, and therefore tend to miss the physical processes involved in radiative transfer processes. Even when data-driven approaches are trained with highly accurate reference data such as ground-based measurements, it is difficult to fully guarantee consistent performance across all spatiotemporal conditions. This study investigates applications of AI in satellite-based air quality monitoring. In particular, the objective is to retain the advantages of AI while preserving the possibility of physical interpretation. The first study introduces AI as a statistical tool for correcting errors in spectral aerosol optical depth (AOD). A simple deep neural network model was trained to learn and correct wavelength-dependent errors in satellite AOD. We used a spectral deconvolution algorithm to physically retrieve the aerosol fine-mode fraction using spectral AODs. The second study replaces a radiative transfer model with an AI-based surrogate model to improve the computational speed and flexibility of a physics-based retrieval algorithm. This approach enhanced the fusion of observations from two satellites and the retrieval of the imaginary refractive index at multiple wavelengths. Finally, an automatized algorithm for methane plume detection was developed using AI-based image segmentation. Introducing AI-based image processing addressed the limitations of geostationary satellites in terms of spatial resolution and sensitivity for methane plume detection

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